Computer Practical

Clouds and Orographic Precipitation

Cloud microphysics in a single-column model

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1 Overview

In this practical you will use a single-column model (SCM) to investigate how cloud microphysics affects precipitation as air is lifted over a hill. You will change one process or environmental condition at a time and interpret the resulting cloud, rain and ice fields.

Keep a practical record: save each figure using a meaningful filename and note the settings and main result.

By the end of the practical you should be able to:

- explain why cloud-droplet number concentration affects the warm-rain process;
- distinguish warm-rain precipitation from precipitation involving the ice phase;
- explain qualitatively how aerosol particles larger than $0.5\,\mu$m influence primary ice in the model;
- describe why a deeper cloud can produce more precipitation;
- recognise that parameterised microphysical processes depend on assumptions as well as on the resolved cloud state.

2 What the model represents

The SCM follows a vertical column of air as it is forced up and over idealised orography. Model time can therefore be interpreted as distance travelled across the hill. Rising air cools, cloud forms and cloud particles may grow into precipitation.

Potential temperature is related to temperature and pressure by

$$\theta = T\left(\frac{10^5}{P}\right)^{R_d/c_p}. \tag{1}$$

A generic model variable ϕ is transported vertically according to

$$\frac{\partial \phi}{\partial t} + w\frac{\partial \phi}{\partial z} = S_\phi, \tag{2}$$

where S_ϕ contains microphysical sources and sinks; falling hydrometeors also sediment relative to the air.

For collection, the mass growth of a large drop can be written schematically as

$$\frac{dm_i}{dt} \propto D_i^{2+b}\sum_j d_j^3 n_j, \tag{3}$$

so changing droplet size and number alters how readily cloud water is converted into rain.

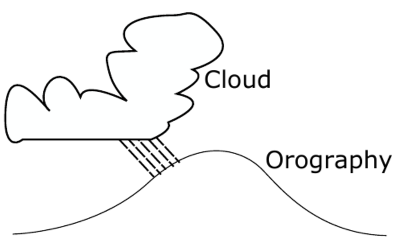

Figure 1: Schematic of the scenario being modelled. Cloud deepens as the air is lifted over the orography and may produce precipitation.

3 Log in, download and compile the code

Use the same SSH/SFTP workflow as in *Tools of the Trade*. On campus, use `130.88.66.57`; replace `YOUR_USERNAME` with your username.

In your SSH window:

```
ssh -Y YOUR_USERNAME@130.88.66.57
```

Open a second terminal window for file transfer:

```
sftp YOUR_USERNAME@130.88.66.57
```

In the SSH window, obtain the model and compile it:

```
git clone https://github.com/EnvModelling/simple-cloud-model
cd simple-cloud-model
make
```

You normally need to compile only once unless the source code changes.

File-transfer reminder. Use SSH for editing/running and SFTP for `get`. Rename files as you download them:

```
get REMOTE_FILE LOCAL_FILENAME
```

4 Inspecting and running the model

Two input files are used in this practical. Open the main model namelist with line numbers displayed:

```
nano -l prac/namelist_prac.in
```

The main parameters are `wr_flag` (about line 24), `num_drop` (28), `tsurf` (43), `t_cbase` (45), `t_ctop` (46), and the ice-process flags (20–22). If the line numbers change, use `Ctrl-W` to search.

Aerosol properties are set in a second file:

```
nano -l prac/namelist.pamm.bam.in
```

Aerosol number concentrations are in `n_aer1` (about line 13); the other aerosol properties are grouped immediately below. The second mode is a pre-configured coarse, dust-like mode used later in an optional experiment.

Run the SCM from the repository root:

```
./run.sh
```

The run writes `/tmp/YOUR_USERNAME/output.nc`. Plot it with:

```
python3 python/scm_plot.py
```

The plot is `/tmp/YOUR_USERNAME/scm_plot.png`. For the default warm-cloud case download it as:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_warm_nd100.png
```

The four panels show cloud-droplet number, cloud water, rain water and ice-crystal number.

Note. Each run overwrites `output.nc` and each plotting run overwrites `scm_plot.png`. Download each figure before the next experiment; later sections reuse earlier reference cases.

To restore just the supplied practical input files after an experiment, use:

```
git restore prac/namelist_prac.in prac/namelist.pamm.bam.in
```

This discards edits to those two files.

5 Parameters used in the practical

Table 1: Main parameters changed in the experiments. Number concentrations in the namelists are expressed per kg of dry air. Near the surface, values such as `100e6` kg^{-1} are of order 100 cm^{-3}.

Parameter	Supplied value	Meaning
`num_drop`	100e6 kg^{-1}	Initial cloud-droplet number concentration
`wr_flag`	.true.	Switch for collision–coalescence / warm-rain formation
`tsurf`	303.15 K	Surface temperature
`t_cbase`	293.15 K	Cloud-base temperature
`t_ctop`	291.15 K	Cloud-top temperature; together with `t_cbase` controls cloud depth
`n_aer1(2)`	0.1e6 kg^{-1}	Background number concentration of the pre-configured coarse dust-like mode
`d_aer1(2)`	2×10^{-6} m	Dry median diameter of the coarse aerosol mode
`mode1_ice_flag`	0	Experimental secondary-ice mode 1 off
`mode2_ice_flag`	0	Experimental secondary-ice mode 2 off
`coll_breakup_flag1`	0	Ice–ice collisional breakup off

Note. The second aerosol mode is the dust-like mode used in Section 6.3. The default `hm_flag=.true.` enables Hallett–Mossop; the three additional SIP options are initially off.

6 Experiments

Save each figure before the next run. Unless stated otherwise, restore the supplied defaults before changing the next parameter. Optional extensions are clearly marked.

6.1 Changing cloud-droplet number

Run the supplied default case and save it as `scm_warm_nd100.png`. Compare panels (a) and (c): as cloud droplets grow and are converted to rain, the droplet field decreases and rain falls towards the surface.

Change `num_drop` (about line 28). The default already provides the `100.e6` reference:

Run	Set `num_drop` to	Save locally as
Reference	`100.e6`	`scm_warm_nd100.png`
Core	`10.e6`	`scm_warm_nd10.png`
Core	`1000.e6`	`scm_warm_nd1000.png`

For the two new runs, download the figures with:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_warm_nd10.png
get /tmp/YOUR_USERNAME/scm_plot.png scm_warm_nd1000.png
```

Prediction. Before running the two additional cases, predict whether increasing droplet number will make collision–coalescence easier or harder when the amount of cloud water is similar.

Exercise. For the default case, where relative to the mountain does rain develop and reach the surface? As cloud-droplet number is reduced or increased, how does the position and amount of rain change? Explain the result in terms of typical droplet size and collision–coalescence.

Note. Optional. Repeat with `num_drop=2000.e6` and save as `scm_warm_nd2000.png`.

6.2 Switching off the warm-rain process

Restore the supplied defaults and turn off warm rain (`wr_flag`, around line 24):

```
wr_flag=.false.
```

Run, plot and save this core case as:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_nowarm_nd100.png
```

Compare directly with `scm_warm_nd100.png`; only the warm-rain process has changed.

Exercise. What happens to the persistence of cloud water and to the rain field when collision–coalescence is disabled? Explain the differences between `scm_warm_nd100.png` and `scm_nowarm_nd100.png` physically.

Note. Optional. If you made `scm_warm_nd2000.png`, repeat it with `wr_flag=.false.`, save as `scm_nowarm_nd2000.png` and compare the pair.

6.3 Cooler conditions: ice, secondary ice and INP

Restore the supplied defaults. The temperature settings are around lines 43–48 of `prac/namelist_prac.in`. Now create a cooler mixed-phase cloud by setting

```
tsurf=290.
t_cbase=270.
t_ctop=265.
```

Run the model with `wr_flag=.true.`, plot it and save it as:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_cold_warmrain.png
```

Then repeat with

```
wr_flag=.false.
```

and save the result as:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_cold_nowarm.png
```

Also examine panel (d). At these temperatures ice-phase precipitation can occur even when warm-rain collision–coalescence is disabled.

Exercise. What happens to precipitation when warm rain is switched off in the cold case? Is the result the same as in Section 6.2? Use the appearance of ice in panel (d) to explain the difference.

Optional extension: adding a pre-configured dust-like coarse aerosol mode

Keep the cold temperatures and restore `wr_flag=.true.`. Leave the three additional SIP flags at their supplied values of zero:

```
mode1_ice_flag=0
mode2_ice_flag=0
coll_breakup_flag1=0
```

The model uses the DeMott primary-ice parameterisation, which depends on temperature and the number of aerosol particles larger than 0.5 μm.

The second aerosol mode is already configured as a coarse (2 μm), weakly hygroscopic dust-like mode. The supplied numbers are:

```
n_aer1(1:3) = 6000.e6, 0.1e6, 0.1e6,
```

Change **only the second number** from `0.1e6` to `10.e6` kg^{-1}:

```
n_aer1(1:3) = 6000.e6, 10.e6, 0.1e6,
```

Run, plot and save this optional case as:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_cold_dust.png
```

Compare with scm_cold_warmrain.png; warm rain remains on and the SIP switches remain off.

Note. This is an idealised perturbation. The DeMott response is driven mainly by the increased number of coarse particles above 0.5 μm.

Exercise. If you complete this optional extension, how does adding the coarse aerosol mode change the ice-crystal number concentration and precipitation field relative to scm_cold_warmrain.png? Relate the response to the model's primary-ice/INP parameterisation.

Optional extension: secondary ice production with the dust mode retained

Keep the cold temperatures, wr_flag=.true., and retain the dust perturbation with n_aer1(2)=10.e6. Now turn on the three additional experimental secondary-ice-production (SIP) options around lines 20–22 of prac/namelist_prac.in:

```
mode1_ice_flag=1
mode2_ice_flag=1
coll_breakup_flag1=2
```

Run, plot and save this optional case as:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_cold_dust_sip.png
```

Option 2 for coll_breakup_flag1 selects the Phillips-type ice–ice breakup treatment. Compare directly with scm_cold_dust.png; only the additional SIP mechanisms have changed.

Exercise. If you complete this optional extension, how much does the ice-crystal number concentration change when the additional SIP mechanisms are enabled? Does the change in ice number alter the rain or cloud-water fields? Why can secondary ice produce many more ice crystals than would be expected from primary ice nucleation alone?

6.4 Deeper cloud

Restore the supplied defaults. Use the existing scm_warm_nd1000.png as the shallow-cloud reference; do not rerun it.

Now change only the cloud depth while keeping num_drop=1000.e6:

```
num_drop=1000.e6
t_cbase=293.15
t_ctop=288.15
```

Run and plot the deeper-cloud case, then save it as:

```
get /tmp/YOUR_USERNAME/scm_plot.png scm_deep_nd1000.png
```

Compare with scm_warm_nd1000.png; droplet number is unchanged, so the comparison isolates cloud depth.

Exercise. How does increasing the cloud depth affect the amount and location of precipitation? Why does giving cloud particles more time and vertical distance in cloud favour precipitation formation?

7 What you should take away

Droplet number controls warm-rain efficiency, while colder clouds can also precipitate through ice-phase pathways. The optional dust and SIP experiments illustrate how parameterised primary and secondary ice processes depend on their assumed controlling variables.

When interpreting a microphysics model, ask which processes are represented, what controls their parameterisations, and what is omitted.